

Srinivas Sivaratri

AI Engineer | LLM evaluation, agent reliability, developer tools

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srinivassivaratri.is-a.dev | Remote, India (IST, UTC+5:30) | Open to full-time and contract

SUMMARY

AI engineer with an MS in Computer Science. Builds LLM systems and the evaluations that check them, and publishes the numbers. Upstream work reviewed by better-auth and go-ethereum maintainers.

SKILLS

AI and LLM: LLM and agent evaluation, guardrails and abstention, RAG, hybrid retrieval (BM25, Reciprocal Rank Fusion, cross-encoder reranking), structured extraction, multimodal models, MCP, Inspect AI, OpenTelemetry

Machine learning: PyTorch, Hugging Face Transformers, PEFT and LoRA, diffusion models, sentence-transformers

Languages: Python, TypeScript, Go, JavaScript, SQL, C

Engineering: FastAPI, Flask, Next.js, React, PostgreSQL, Redis, SQLite, Docker, GitHub Actions, CI/CD, pytest, Vitest

EXPERIENCE

Software Engineer, Independent | Remote

May 2025 to Present

- Built eight AI projects with public repos and published results, including losses to baselines.
- Traced an email-enumeration leak in better-auth to its Have I Been Pwned plugin and proposed a request-scoped skip flag with a regression test. Maintainers adopted the approach and shipped it (PR #5998).
- Designed "geth db truncate-freezer" for go-ethereum (PR #31351), revised across three core-maintainer review rounds to batch-process instead of holding about 15 GB in memory. The core team later built its own version.

Software Engineer, Accenture | Bangalore, India

Jul 2021 to Jul 2022

- Wrote a Python ETL script that automated data processing, saved 50 labor hours per month, and cut errors.
- Implemented a GitHub Actions CI/CD pipeline.

PROJECTS

Bouncer | Python, OpenTelemetry

github.com/sivaratrisrinivas/bouncer-eval

- Built an eval for when an AI agent with write access to a store may issue refunds. Each case has five possible actions (ACT, REPLY, ASK, ESCALATE, ABSTAIN), and grading is deterministic with no LLM judges.
- On 50 cases, a rules baseline had a 2.0% Unsafe Action Rate. gemma-4-31b and gpt-oss-120b both had 10.0%.

llmfr (LLM Flight Recorder) | Python, Hugging Face, SQLite

github.com/sivaratrisrinivas/llm-flight-recorder

- Built a CLI that records a local LLM generation token by token, replays it, and names the first step where two runs diverge, so later differences are tagged as fallout. The OpenAI adapter fails closed when logprobs are missing.
- On 30 Qwen2.5-0.5B-Instruct prompts, runs disagreed in 10 cases under sampling changes and 13 under decoding changes.

POSTDATED | TypeScript, Next.js, Cerebras, Vitest

github.com/sivaratrisrinivas/postdated

- Selected as one of about 150 builders from 5,000+ applications for the five-hour Push to Prod hackathon in Bengaluru.
- Built an app that turns photos of hospital discharge paperwork into a predicted, itemized insurance rejection, with one document request or doctor question per fixable line. Kept multimodal extraction apart from deterministic policy arithmetic and a guard that blocks unsupported medical claims.
- Added 38 tests on the money and guard paths, a 40-case extraction eval harness, and offline CI gates.

Bingo | Python, sentence-transformers, NLTK

github.com/sivaratrisrinivas/Bingo

- Implemented BM25 from scratch and compared it with semantic search, Reciprocal Rank Fusion, and a cross-encoder reranker. On 10 queries over 5,000 movies, BM25 led at 0.38 Precision@5 (semantic 0.32, hybrid reranking 0.36).

SokoFlow | Python, PyTorch

github.com/sivaratrisrinivas/sokoflow

- Trained a conditional diffusion policy (CNN board encoder, Transformer denoiser) to generate Sokoban plans. Solved 15 of 240 scrambled 8x8 boards (6.2%) against 94.2% for a BFS baseline, and published the gap.

Threadmark and context-CI | In progress

github.com/sivaratrisrinivas/threadmark

- Threadmark: a coding agent that logs to SQLite and, after a crash, reconciles with the live Git tree before it writes.
- context-CI: a CLI spec and proof testing whether one MCP tool change alters a coding agent's results (Inspect AI).

EDUCATION

MS, Computer Science, University of Alabama at Birmingham, AL, USA

Aug 2023 to May 2025

BS, Computer Science, BV Raju Institute of Technology, Telangana, India

Aug 2017 to Jun 2021

Boot.dev: HTTP Servers in Go, RAG, CI/CD with GitHub Actions and Docker, Memory Management in C, SQL